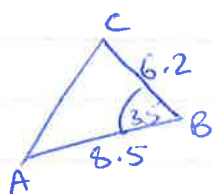


October 2020 Pure 8 Mech

1)



$$\frac{1}{2} \times 8.5 \times 6.2 \times \sin 35$$

$$= 15.1 \text{ cm}^2 //$$

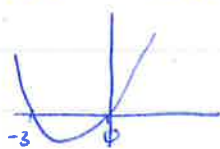
2)

$$e^x \rightarrow e^{x+3} \rightarrow e^{2x+3}$$

• translate $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$, then

• stretch s.f $\frac{1}{2}$ parallel to x-axis //

3a) $f(x) = 2x(x+3)$



$$f(x) = 2[x^2 + 3x]$$

$$= 2\left[\left(x + \frac{3}{2}\right)^2 - \frac{9}{4}\right]$$

$$= 2\left(x + \frac{3}{2}\right)^2 - \frac{9}{2}$$

$$\therefore f(x) \geq -\frac{9}{2} //$$

b) it is not a one-to-one mapping $\therefore$ it has no inverse //

c) $g^{-1}(x) = \frac{x-2}{3}$

$$fg(-2) = f(-4) = 8$$

$$\therefore \frac{x-2}{3} = 8$$

$$x = 26 //$$

d) $2x^2 + 6x > 3x + 2$

$$2x^2 + 3x - 2 > 0$$

$$(2x-1)(x+2) > 0$$

$$x < -2 \text{ or } x > \frac{1}{2}$$

$$\{x: x < -2\} \cup \{x: x > \frac{1}{2}\} //$$



4a) p.o.i $\Rightarrow \frac{d^2y}{dx^2} = 0$

$$\frac{dy}{dx} = \frac{2}{k-3x} x - 3 + 2x - 3 = \frac{-6}{k-3x} + 2x - 3 = -6(k-3x)^{-1} + 2x - 3$$

$$\frac{d^2y}{dx^2} = -18(k-3x)^{-2} + 2 //$$



when $x=1$, $\frac{d^2y}{dx^2} = 0$

$$-18(k-3)^{-2} + 2 = 0$$

$$2 = \frac{18}{(k-3)^2}$$

$$(k-3)^2 = 9$$

$$k-3 = 3 \text{ or } k-3 = -3$$

$$k = 6 \quad k = 0$$

k is positive ($k > 0$) so $k = 6$.

b) $0 = 2\ln(k-3x) + x^2 - 3x = f(x)$

$$f(0.5) = 2\ln(6-1.5) + 0.5^2 - 3 \times 0.5 = 1.758 > 0$$

$$f(1.5) = 2\ln(6-4.5) + 1.5^2 - 3 \times 1.5 = -1.439 < 0$$

$\therefore$ there is a sign change

$\therefore$ there is a root between 0.5 and 1.5

c) $f(x) = 2\ln(6-3x) + x^2 - 3x$

$$f'(x) = \frac{2x-3}{6-3x} + 2x-3 = \frac{-6}{6-3x} + 2x-3$$

$$x_0 = 1$$

$$x_1 = 1 - \frac{2\ln(6-3) + 1-3}{\frac{-6}{3} + 2-3} = 1.0657415$$

$$x \approx 1.06568$$

$$f(1.065685) = \dots < 0$$

$$f(1.065675) = \dots > 0$$

5a) $\frac{dy}{dx} = \frac{dy/db}{dx/db} = \frac{3t^2 e^{-2t} + 2e^{-2t} t^3}{-3t^{-2}} = 0$

$$e^{-2t}(3t^2 - 2t^3) = 0$$

$$e^{-2t} = 0 \text{ no solutions since } e^{-2t} > 0$$

$$3t^2 - 2t^3 = 0$$

$$t^2(3-2t) = 0$$

$$t = 0 \text{ not valid since } t > 0$$

or $t = 1.5$

$$\therefore (2, \frac{27}{8}e^{-3})$$

$$b) \quad x=1 \quad t=3$$

$$x=6 \quad t=0.5$$

$$\int_1^6 y \frac{dx}{dt} = \int_3^{0.5} t^3 e^{-2t} \cdot -3t^{-2} dt = \int_3^{0.5} -3te^{-2t} dt = \int_3^{0.5} 3te^{-2t} dt //$$

$$c) \quad u=3t \quad v=-\frac{1}{2}e^{-2t}$$

$$\frac{du}{dt} = 3 \quad \frac{dv}{dt} = e^{-2t}$$

$$-\frac{3}{2}te^{-2t} + \frac{3}{2} \int e^{-2t} dt$$

$$= \left[-\frac{3}{2}te^{-2t} + \frac{3}{4}e^{-2t} \right]_{0.5}^3$$

$$= \left(-\frac{9}{2}e^{-6} - \frac{3}{4}e^{-6} \right) - \left(-\frac{3}{4}e^{-1} - \frac{3}{4}e^{-1} \right)$$

$$= -\frac{21}{4}e^{-6} + \frac{3}{2}e^{-1} //$$

$$6a) \quad 4x \frac{dy}{dx} + 4y = 4x + 16y \frac{dy}{dx} - 9$$

$$4x \frac{dy}{dx} - 16y \frac{dy}{dx} = 4x - 9 - 4y$$

$$\frac{dy}{dx} (4x - 16y) = 4x - 4y - 9$$

$$\frac{dy}{dx} = \frac{4x - 4y - 9}{4x - 16y}$$

$$b) \quad \text{parallel to y-axis: } 4x - 16 = 0$$

$$x = 4y \quad \text{~~parallel to x-axis~~}$$

$$\text{parallel to x-axis: } 4x - 4y - 9 = 0$$

$$\frac{4x - 9}{4} = y$$

when $x = 4y$

$$16y^2 = 2(16y^2 + 4y^2) - 36y$$

$$16y^2 = 32y^2 + 8y^2 - 36y$$

~~16y^2 = 32y^2 + 8y^2 - 36y~~

$$0 = 24y^2 - 36y$$

$$0 = 12y(2y - 3)$$

$$y = 0 \text{ or } y = \frac{3}{2}$$

$$P(6, 3/2)$$

$$4x \left(\frac{4x-9}{4} \right) = 2x^2 + 8 \left(\frac{4x-9}{4} \right)^2 - 9x$$

$$4x^2 - 9x = 2x^2 + 8 \left(\frac{16x^2 - 72x + 81}{16} \right) - 9x$$

$$4x^2 - 9x = 2x^2 + 8x^2 - 36x + \frac{81}{2} - 9x$$

$$0 = 6x^2 - 36x + \frac{81}{2}$$

$$0 = 12x^2 - 72x + 81$$

$$0 = 4x^2 - 24x + 27$$

$$0 = (2x - 9)(2x - 3)$$

$$x = \frac{9}{2} \text{ or } x = \frac{3}{2}$$

$$y = 225 \quad y = -3/4$$

$$\therefore Q\left(\frac{3}{2}, -\frac{3}{2}\right)$$

$$PQ = \sqrt{\left(\frac{3}{2} - 6\right)^2 + \left(-\frac{3}{2} - \frac{3}{2}\right)^2}$$

$$= \sqrt{\frac{81}{4} + \frac{81}{4}}$$

$$= \frac{9}{4} \sqrt{5}$$

$$\begin{aligned}
 7a) \quad s & \quad v = u + at \\
 u &= \begin{pmatrix} 7 \\ 6 \end{pmatrix} \quad v = \begin{pmatrix} 7 \\ 6 \end{pmatrix} + 3 \begin{pmatrix} -4 \\ 2 \end{pmatrix} \\
 v &= ? \\
 a &= \begin{pmatrix} -4 \\ 2 \end{pmatrix} \quad v = \begin{pmatrix} -5 \\ 12 \end{pmatrix} \\
 t &= 3
 \end{aligned}$$

$$\therefore \text{speed} = \sqrt{25 + 144} = 13 \text{ ms}^{-1} //$$

$$\begin{aligned}
 b) \quad s &= ut + \frac{1}{2} at^2 \\
 s &= 3 \begin{pmatrix} 7 \\ 6 \end{pmatrix} + \frac{9}{2} \begin{pmatrix} -4 \\ 2 \end{pmatrix} \\
 s &= \begin{pmatrix} 3 \\ 27 \end{pmatrix} = 3\mathbf{i} + 27\mathbf{j} \text{ m} //
 \end{aligned}$$

$$8a) \quad c = 8$$

$$b) \quad \frac{dv}{dt} = 2at + b$$

$$-0.12 = 2a(5) + b$$

$$-0.12 = 10a + b \quad \text{--- (1)}$$

$$2.96 = a \times 18^2 + b \times 18 + 8$$

$$-5.04 = 324a + 18b$$

$$-0.28 = 18a + b \quad \text{--- (2)}$$

$$\therefore -10a - 0.12 = -0.28 - 18a$$

$$8a = -0.16$$

$$a = -0.02 //$$

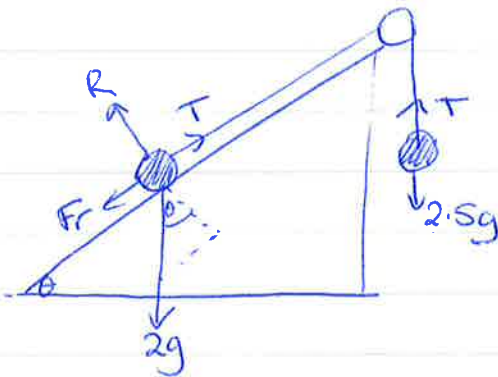
$$\therefore b = 0.08 //$$

$$c) \quad s = \int_0^{18} -0.02t^2 + 0.08t + 8 \, dt$$

$$= \left[-\frac{0.02}{3} t^3 + 0.04t^2 + 8t \right]_0^{18}$$

$$= \frac{-0.02}{3} \times 18^3 + 0.04 \times 18^2 + 8 \times 18 - 0 = 118.08 \approx 118 \text{ m} //$$

9



$$\cos \theta = 4/5$$

$$\sin \theta = 3/5$$

a) $s = 1.5$

$$u = 0$$

$$v = 1.2$$

$$a = ?$$

$$v^2 = u^2 + 2as$$

$$1.2^2 = 0 + 2 \times a \times 1.5$$

$$a = \frac{1.2^2}{2 \times 1.5} = 0.48 \text{ ms}^{-2} //$$

b) ($\downarrow$) $2.5g - T = 2.5 \times 0.48$

$$T = 2.5g - 2.5 \times 0.48$$

$$T = 23.3 \text{ N} //$$

c) ($\uparrow$) $R - 2g \cos \theta = 0$

$$R = 2g \times \frac{4}{5}$$

$$R = 15.68 \text{ N}$$

($\rightarrow$) $T - Fr - 2g \sin \theta = 2 \times 0.48$

$$23.3 - Fr - 11.76 = 0.96$$

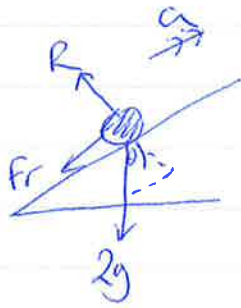
$$Fr = 10.58 \text{ N}$$

$$\boxed{Fr = \mu R}$$

$$10.58 = \mu \times 15.68$$

$$\mu = 0.675 // \text{ (3sf)}$$

d) new:



~~10.58~~

$F_r = 10.58$ (still at its max)

$$(\rightarrow) -2g \sin \theta - 10.58 = 2a$$

$$- \frac{11.76 - 10.58}{2} = a$$

$s = ?$

$a = -11.17$

$u = 1.2$

~~1.2~~

$v = 0$

$v^2 = u^2 + 2as$

$a = -11.17$

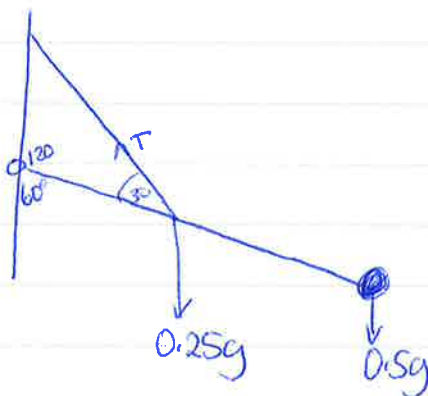
$0 = 1.2^2 + 2(-11.17)s$

$t = x$

$s = \frac{1.2^2}{2 \times 11.17}$

$s = 0.0645 \text{ m} //$ (3sf)

10)



(2) $0.25g \times 0.4 \sin 60 + 0.5 \times 0.8 \sin 60$

$- T \times 0.4 \sin 30 = 0$

$T = 21.2 \text{ N} //$

b) at A assum



(1) $Y + T \cos 30 - 0.25g - 0.5g = 0$

~~11.00973~~ $Y = -11.00973$

(2) $X - T \sin 30 = 0$

$X = 10.6$

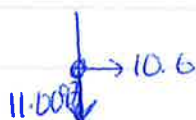
Q10bii) $\tan \theta = \frac{11.0097}{10.6}$

$\theta = 46.1$

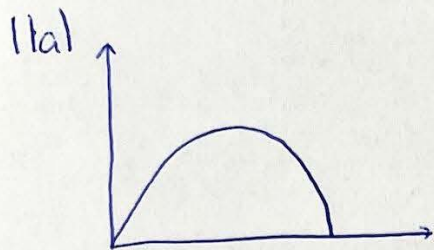
below horizontal



so at A



$\therefore \text{force} = \sqrt{11.0097^2 + 10.6^2}$
 $= 15.3 \text{ N} //$



(→) S
 $u = u$
 $v = u$
 $a = 0$
 t

(↑) $S = 0$
 $u = v$
 v
 $a = -g$
 t

OC

(→) $S = ut + \frac{1}{2}at^2$

$OC = ut$

(↑) $S = ut + \frac{1}{2}at^2$

$0 = vt - \frac{1}{2}gt^2$

$0 = t(v - \frac{1}{2}gt)$

$t = 0$ or $v = \frac{1}{2}gt$

$\frac{2v}{g} = t$

$\therefore OC = \frac{2vu}{g} //$

b) O to A

(→) velocity is always $u //$

(↑) $S = h$
 $u = v$
 $v = ?$
 $a = -g$
 t

$v^2 = u^2 + 2as$

velocity = $\sqrt{v^2 - 2gh} //$

c) A to B from A start to B end

(→) $S = d$
 $u = u$
 $v = u$
 $a = 0$
 t

$S = ut + \frac{1}{2}at^2$
 $d = ut$
 $t = \frac{d}{u}$

(↑) $S = 0$
 $u = \sqrt{v^2 - 2gh} \rightarrow \text{velocity at A!!!}$
 $v = \text{velocity at B}$
 $a = -g$
 $t = \frac{d}{u}$

~~$S = ut + \frac{1}{2}at^2$~~

~~$0 = vt - \frac{1}{2}gt^2$~~

$S = ut + \frac{1}{2}at^2$

$0 = t\sqrt{v^2 - 2gh} - \frac{1}{2}gt^2$

~~$0 = vt - \frac{1}{2}gt^2$~~

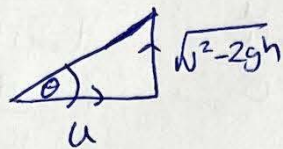
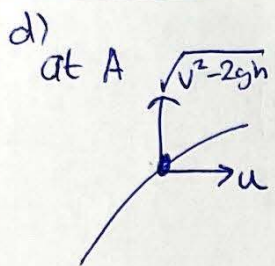
$0 = t(\sqrt{v^2 - 2gh} - \frac{1}{2}gt)$

$t = 0$ or $\sqrt{v^2 - 2gh} = \frac{1}{2}gt$

sub $t = \frac{d}{u}$

$\sqrt{v^2 - 2gh} = \frac{gd}{2u}$

$\frac{2u}{g}\sqrt{v^2 - 2gh} = d //$



$$\tan \theta = \frac{\sqrt{v^2 - 2gh}}{u}$$

$$\frac{1}{2} = \frac{\sqrt{v^2 - 2gh}}{u}$$

$$u = 2\sqrt{v^2 - 2gh}$$

from (b)

$$d = \frac{2u}{g} \sqrt{v^2 - 2gh}$$

$$\frac{gd}{2\sqrt{v^2 - 2gh}} = u$$

$$\therefore 2\sqrt{v^2 - 2gh} = \frac{gd}{2\sqrt{v^2 - 2gh}}$$

$$4(v^2 - 2gh) = gd$$

$$v^2 - 2gh = \frac{gd}{4}$$

$$v^2 = \frac{gd}{4} + 2gh$$

$$v = \sqrt{\frac{gd}{4} + 2gh} //$$

MS version

$$= \sqrt{\frac{gd + 8gh}{4}}$$

$$= \sqrt{\frac{1}{4}} \sqrt{gd + 8gh}$$

$$= \frac{1}{2} \sqrt{gd + 8gh}$$